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HW-5

1. $\widehat{rent} = 6 + 0.035\,area + 4.5 \cdot 1\{furnish\} + 0.012\,area \cdot 1\{furnish\}$

Yes

$= 6 + 0.035(800) + 4.5 + 0.012(800)$

$= 6 + 28 + 4.5 + 9.6 = 48.1$

Rent = 48,100

2. $U = 1 - F \Rightarrow F = 1 - U$

$\hat{y} = 6 + 0.035A + 4.5 \cdot 1(1-U) + 0.012A(1-U)$

$= 10.5 + 0.047A - 4.5U - 0.012U$

all coefficients Δ, but none of the fitted values

3. None is wrong. 1 estimates the marginal associa^{n} btw. fertilizer & yield w/o control on soil quality. 2 estimates the partial association while holding soil quality constant.

4. The single coefficient estimates a pooled common slope than channel's true slope. It understates online & overstates online ad.

5. clay, loam, sand → dummy var trap or perfect multicollinearity

With an intercept, a 3-level categorical var. req. only 3-1 = 2 dummy cols., the omitted category becomes the ref. If 3 are there → leads to multi collinearity & non-unique OLS coffients

6.

	L	W	
A	32	48	(+12)
B	44	38	(-6)

Single universal effect is misleading. The effect +16 for A & −6 for B.

7. coding assumes equally spaced seriority levels & a linear, constant per level effect on response.

8. OLS fitted values & RSS remains unaffected by units as coefficients rescale accordingly.
GD is affected as feature scaling Δs gradient magnitudes & the condition of the optimizn problem.

9. OLS → standardizn only reparameterizes the predictors & coeff. w/o Δing the fitted values, as OLS minimizes the residual sum of sq. w/o a coeff penalty.

Ridge → stand. Δs the magni. of coeff. in L2, thus, Δ in the amt. of shrinkage applied to each predictor & Δs fitted values.

10. 0 training error with K=1 shows memorizn of training obs., not good generalizn.
K = n = 500, predicns become global in high bias & low var.

11. Missing → out of sample evaluan using an indep., appropriate time ordered time set. The quoted train err. will be too small with true future predicn error.

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12. $p >> n \Rightarrow RSS = 0$
1. risk of overfit
2. non-unique & unstable coeff.

13. Collinearity is tolerable when the obj. is predic^n on data with similar predictor relationship because fitted values will remain stable. But if,
the obj. is to estimate or interpret indi. predictor effects, like determining the sep. contribu^n of 2 highly correlated fertilizer.

14. Analyst violates the rule that all data-dep. processing must be learnt from the training fold only.
Full dataset standardiza^n creates data-leakage & can prod. an overly optimistic (too-small CV error estimate.

15. 2 fold CV will have larger error as each model is trained on 50% of the obs. as in 90% in 10-fold CV. But ordering is not guaranteed for a particular dataset.